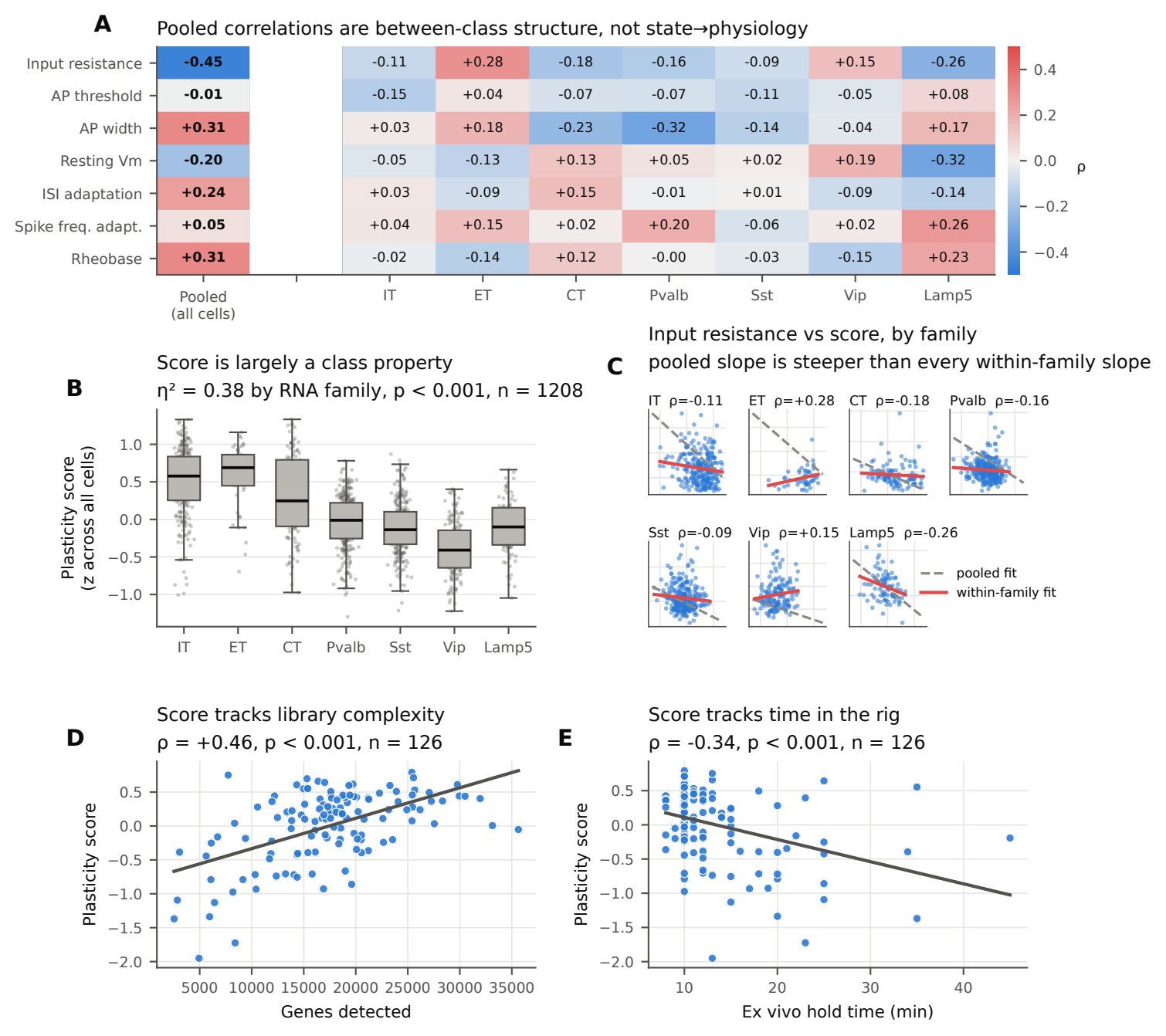

A
Pooled correlations are between-class structure, not state→physiology
Input resistance
AP threshold
AP width
Resting Vm
ISI adaptation
Spike freq. adapt.
Rheobase
-0.45
-0.01
+0.31
-0.20
+0.24
+0.05
+0.31
-0.11 +0.28 -0.18 -0.16 -0.09 +0.15 -0.26
-0.15 +0.04 -0.07 -0.07 -0.11 -0.05 +0.08
+0.03 +0.18 -0.23 -0.32 -0.14 -0.04 +0.17
-0.05 -0.13 +0.13 +0.05 +0.02 +0.19 -0.32
+0.03 -0.09 +0.15 -0.01 +0.01 -0.09 -0.14
+0.04 +0.15 +0.02 +0.20 -0.06 +0.02 +0.26
-0.02 -0.14 +0.12 -0.00 -0.03 -0.15 +0.23
Pooled (all cells)
IT
ET
CT
Pvalb
Sst
Vip
Lamp5
0.4
0.2
0.0
−0.2
−0.4
ρ
B
Score is largely a class property
η² = 0.38 by RNA family, p < 0.001, n = 1208
Plasticity score (z across all cells)
1.0
0.5
0.0
−0.5
−1.0
IT
ET
CT
Pvalb
Sst
Vip
Lamp5
C
Input resistance vs score, by family
pooled slope is steeper than every within-family slope
IT ρ=-0.11
ET ρ=+0.28
CT ρ=-0.18
Pvalb ρ=-0.16
Sst ρ=-0.09
Vip ρ=+0.15
Lamp5 ρ=-0.26
pooled fit
within-family fit
D
Score tracks library complexity
ρ = +0.46, p < 0.001, n = 126
Plasticity score
0.5
0.0
−0.5
−1.0
−1.5
−2.0
5000 10000 15000 20000 25000 30000 35000
Genes detected
E
Score tracks time in the rig
ρ = -0.34, p < 0.001, n = 126
Plasticity score
0.5
0.0
−0.5
−1.0
−1.5
−2.0
10 20 30 40
Ex vivo hold time (min)